

EXP 21:

Aim

To write a Pandas program to swap the uppercase and lowercase characters of a specified column in a given DataFrame.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing a character/string column.
3. Select the specified column.
4. Use the swapcase() function to change uppercase letters to lowercase and lowercase letters to uppercase.
5. Display the original and modified DataFrame.
6. Stop.

Code

```
import pandas as pd

# Create a DataFrame

data = {

    "Name": ["Sabiha", "PYTHON", "DataScience", "Hello World"]

}

df = pd.DataFrame(data)

print("Original DataFrame:")

print(df)

# Swap the cases of the specified column

df["Name"] = df["Name"].str.swapcase()
```

```
print("\nDataFrame after swapping cases:")
```

```
print(df)
```

Output:

```
Original DataFrame:
```

```
      Name
0   Sabiha
1   PYTHON
2 DataScience
3 Hello World
```

```
DataFrame after swapping cases:
```

```
      Name
0  sABIHA
1  python
2 dATAsCIENCE
3 hELLO WORLD
```

Result

The cases of the characters in the specified **Name** column are successfully swapped.

EXP 22 :

Aim

To write a Python program to draw a line graph with suitable labels for the **x-axis, y-axis, and title.**

Algorithm

1. Import the matplotlib.pyplot library.
2. Define the x-axis and y-axis values.
3. Plot the values using plt.plot().
4. Add a label to the x-axis using plt.xlabel().
5. Add a label to the y-axis using plt.ylabel().
6. Add a title using plt.title().
7. Display the graph using plt.show().

Code

```
import matplotlib.pyplot as plt
```

```
# Values for x and y axis
```

```
x = [0, 10, 20, 30, 40, 50]
```

```
y = [0, 30, 60, 90, 120, 150]
```

```
# Draw the line
```

```
plt.plot(x, y)
```

```
# Add labels and title
```

```
plt.xlabel("x - axis")
```

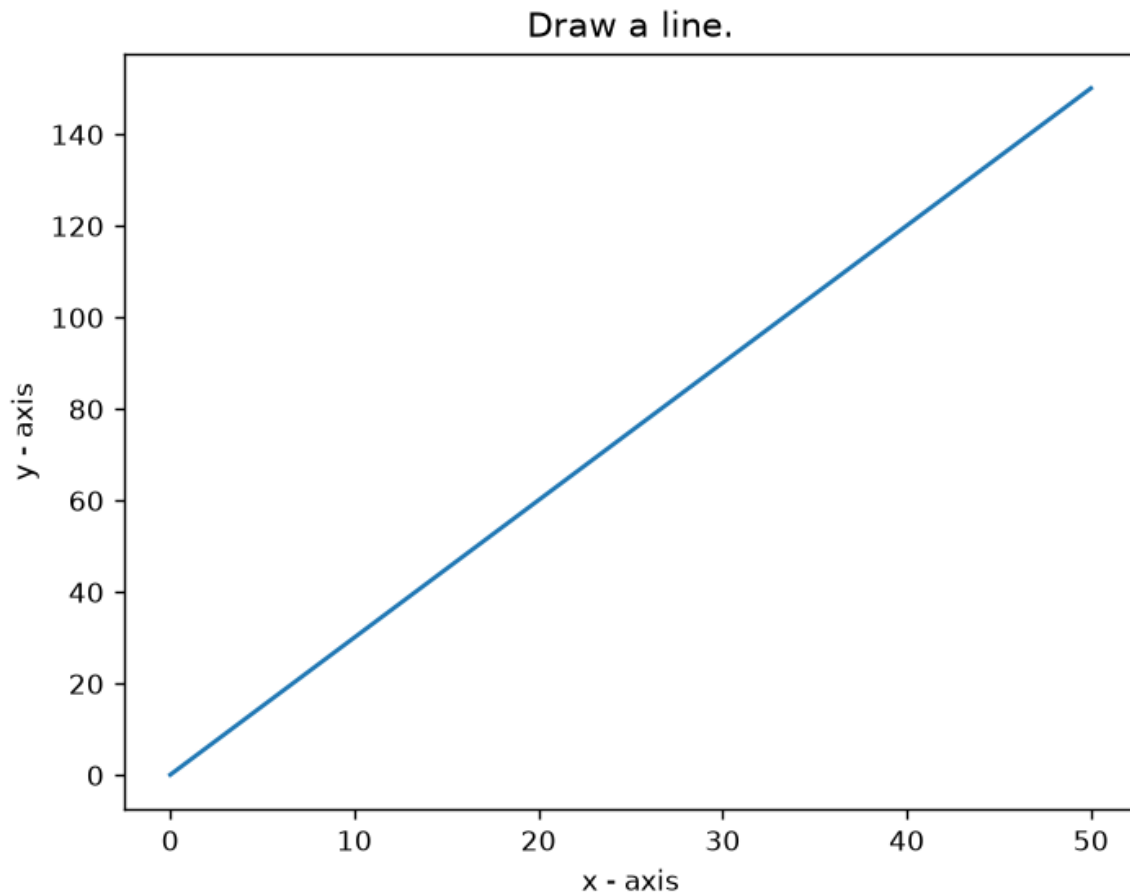
```
plt.ylabel("y - axis")
```

```
plt.title("Draw a line.")
```

```
# Display the graph
```

```
plt.show()
```

Output:



Result

The Python program was executed successfully, and a **line graph with suitable x-axis label, y-axis label, and title** was displayed as shown in the given figure.

EXP 23:

Aim

To write a Python program to read x-axis and y-axis values from a text file and draw a line with suitable labels and a title.

Test Data — test.txt

1 2

2 4

3 1

Algorithm

1. Import the matplotlib.pyplot library.
2. Open the test.txt file in read mode.
3. Read the x-axis and y-axis values from the file.
4. Store the values in separate lists.
5. Plot the x and y values using plt.plot().
6. Add labels for the x-axis and y-axis.
7. Add a suitable title.
8. Display the graph.

Code

```
import matplotlib.pyplot as plt
```

```
x = []
```

```
y = []
```

```
# Read values from text file
```

```
with open("test.txt", "r") as file:
```

```
for line in file:
```

```
    values = line.split()
```

```
    x.append(float(values[0]))
```

```
    y.append(float(values[1]))
```

```
# Draw the line
```

```
plt.plot(x, y)
```

```
# Add labels and title
```

```
plt.xlabel("x - axis")
```

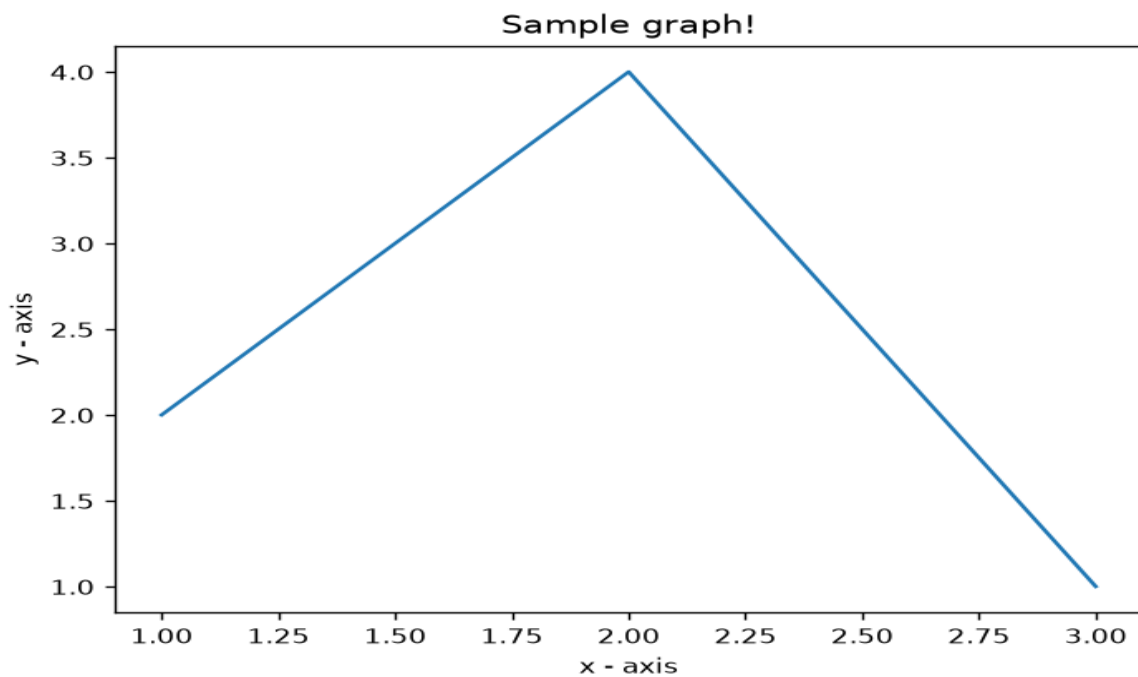
```
plt.ylabel("y - axis")
```

```
plt.title("Sample graph!")
```

```
# Display the graph
```

```
plt.show()
```

Output:



Result

The program was executed successfully. The x-axis and y-axis values were read from **test.txt**, and a line graph with suitable labels and title was displayed.

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